

Coordinate Geometry

Chapter 7 - Hinglish Notes, Formulas, Examples + Solutions

Ye Chapter 7 (Coordinate Geometry) ka complete guide hai. Pehle red-page ke core basics (axis, coordinates, quadrants, square/sqrt, average, ratio), fir chapter ka main content - distance formula, section formula aur midpoint. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. X-axis, Y-axis, Origin

EXAM * Exam me CONSTANTLY aata hai

- X-axis = leti (horizontal) line; Y-axis = khadi (vertical) line.
- Dono milte hain ORIGIN (0, 0) pe.

2. Coordinates (x, y)

EXAM * Exam me CONSTANTLY aata hai

- Point ko (x, y) likhte hain.
- x = abscissa (X par doori), y = ordinate (Y par doori).
- Order important hai: (3, 5) != (5, 3).

3. Signs / Quadrants

EXAM * Exam me CONSTANTLY aata hai

- Q1 (+,+), Q2 (-,+), Q3 (-,-), Q4 (+,-).
- X-axis pe y = 0; Y-axis pe x = 0.

4. Square aur Square Root

- $a^2 = a \times a$. $\sqrt{25} = 5$.
- Distance formula me squares aur sqrt use hote hain.

5. Negative ka Square

- $(-4)^2 = 16$ (negative ka square POSITIVE hota hai).
- Isliye distance hamesha positive aati hai.

6. Average (Mean)

- Do numbers ka average = $(a + b)/2$.
- Midpoint formula me bas average lena hota hai.

7. Ratio (m1 : m2)

- Ratio = do hisson ki tulna.
- Section formula me point segment ko m1 : m2 me baant-ta hai.

PART B - Chapter 7 ka Main Content

8. Distance Formula

EXAM * Exam me CONSTANTLY aata hai

Do points ke beech ki doori:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Points: A(x₁, y₁) aur B(x₂, y₂).

- Example: (0,0) aur (3,4) ki doori = $\sqrt{9+16} = \sqrt{25} = 5$.

9. Distance from Origin

EXAM * Exam me CONSTANTLY aata hai

Origin (0,0) se point (x, y) ki doori:

- Distance = $\sqrt{x^2 + y^2}$.
- Example: (6, 8) ki origin se doori = $\sqrt{36+64} = \sqrt{100} = 10$.

10. Section Formula

EXAM * Exam me CONSTANTLY aata hai

Point P jo A(x₁,y₁) aur B(x₂,y₂) ko m₁:m₂ me baant-ta hai:

$$P = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

- Yaad: ratio m₁:m₂ me 'door wale' point ka coordinate pehle aata hai.

11. Midpoint Formula

EXAM * Exam me CONSTANTLY aata hai

Beech (mid) ka point (ratio 1:1):

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- Bas dono x ka average aur dono y ka average le lo.
- Example: (2,3) aur (4,7) ka midpoint = (3, 5).

12. Collinear Points

- Agar teen points ek hi seedhi line par hon to wo COLLINEAR kehlate hain.
- Check: AB + BC = AC ho (distances jod kar) to points collinear hain.

13. Word Problems / Uses

- Diagonal ya side ki length nikalna -> distance formula.
- Beech ka point -> midpoint; ratio me baant -> section formula.
- Type of figure (square/rhombus) check karna -> sides/diagonals compare.

Quick Revision

Distance = $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$.

From origin = $\sqrt{x^2 + y^2}$.

Section: $((m_1x_2+m_2x_1)/(m_1+m_2), (m_1y_2+m_2y_1)/(m_1+m_2))$.

Midpoint = $((x_1+x_2)/2, (y_1+y_2)/2)$.

Quadrants: Q1(+,+) Q2(-,+) Q3(-,-) Q4(+,-).

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1 (Distance): A(1,2) aur B(4,6) ki doori?

- Step: Distance = $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$.
- Step: = $\sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9 + 16}$.
- Step: = $\sqrt{25}$.

Answer:Distance = 5

Example 2 (From origin): Point (6, 8) ki origin se doori?

- Step: Distance = $\sqrt{x^2 + y^2} = \sqrt{36 + 64}$.
- Step: = $\sqrt{100}$.

Answer:10

Example 3 (Midpoint): (2,3) aur (4,7) ka midpoint?

- Step: Midpoint = $((x_1+x_2)/2, (y_1+y_2)/2)$.
- Step: = $((2+4)/2, (3+7)/2) = (6/2, 10/2)$.

Answer:(3, 5)

Example 4 (Midpoint): (-2, 4) aur (6, -2) ka midpoint?

- Step: $x = (-2+6)/2 = 2$; $y = (4 + (-2))/2 = 1$.

Answer:(2, 1)

Example 5 (Section): A(1,2), B(4,5) ko 1:2 me baant ne wala point?

- Step: $m_1=1, m_2=2. x = (1x_4 + 2x_1)/(1+2) = (4+2)/3 = 2$.
- Step: $y = (1x_5 + 2x_2)/3 = (5+4)/3 = 3$.

Answer:(2, 3)

Example 6 (Quadrant): Point (-3, 5) kis quadrant me hai?

- Step: x negative, y positive -> (-,+).

Answer:Quadrant 2 (Q2)

Example 7 (Distance check): Kya (0,0),(3,4) ki doori 5 hai?

- Step: $\sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25}$.

Answer:Haan, 5

Example 8 (Equal distance): (0,0) se (5,0) aur (0,5) - doori barabar?

- Step: (5,0): $\sqrt{25} = 5$. (0,5): $\sqrt{25} = 5$.

Answer:Haan, dono 5

Example 9 (Midpoint reverse): Midpoint (3,4), ek end (1,2). Doosra end?

- Step: $(1+x)/2 = 3 \rightarrow x = 5$; $(2+y)/2 = 4 \rightarrow y = 6$.

Answer:(5, 6)

Example 10 (Collinear idea): (1,1),(2,2),(3,3) ek line par?

- Step: Teeno me $y = x$ pattern hai, ek hi seedhi line.

Answer:Haan, collinear

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1. Origin ke coordinates kya hote hain?
- Q2. Point (x, y) me x aur y ko kya kehte hain?
- Q3. Distance formula kya hai?
- Q4. Midpoint formula kya hai?
- Q5. (0,0) aur (3,4) ki doori kya hai?
- Q6. (6,8) ki origin se doori kya hai?
- Q7. (2,4) aur (6,8) ka midpoint?
- Q8. Point (-5, -2) kis quadrant me hai?
- Q9. X-axis par kisi point ki y-coordinate kya hoti hai?
- Q10. Y-axis par kisi point ki x-coordinate kya hoti hai?
- Q11. (1,1) aur (5,5) ki doori?
- Q12. Section formula kis kaam aata hai?
- Q13. (-2,3) aur (4,3) ka midpoint?
- Q14. Quadrant 3 me signs kaise hote hain?
- Q15. (0,0) aur (0,7) ki doori?
- Q16. Midpoint (4,5), ek end (2,3) -> doosra end?
- Q17. (3,0) aur (0,4) ki doori?
- Q18. A(2,3), B(8,3) ko 1:1 me baant ne wala point?
- Q19. (7, -2) kis quadrant me hai?
- Q20. Collinear points ka matlab kya hai?

Step-by-Step Solutions

- Q1. Origin ke coordinates?
 - Step: Jahan dono axis milte hain.**Answer:**(0, 0)
- Q2. x aur y ko kya kehte?
 - Step: X par doori = abscissa, Y par = ordinate.**Answer:**abscissa aur ordinate
- Q3. Distance formula?
 - Step: Do points ke beech.**Answer:** $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$
- Q4. Midpoint formula?
 - Step: Dono coordinates ka average.**Answer:** $((x_1+x_2)/2, (y_1+y_2)/2)$
- Q5. (0,0),(3,4) doori?
 - Step: $\sqrt{9+16} = \sqrt{25}$.

Answer:5

Q6. (6,8) origin se doori?

- Step: $\sqrt{36+64} = \sqrt{100}$.

Answer:10

Q7. (2,4),(6,8) midpoint?

- Step: $((2+6)/2, (4+8)/2) = (4, 6)$.

Answer:(4, 6)

Q8. (-5,-2) quadrant?

- Step: x-, y- $\rightarrow$ (-,-).

Answer:Q3

Q9. X-axis par y?

- Step: X-axis par height 0 hoti hai.

Answer:0

Q10. Y-axis par x?

- Step: Y-axis par doori 0 hoti hai.

Answer:0

Q11. (1,1),(5,5) doori?

- Step: $\sqrt{16+16} = \sqrt{32} = 4\sqrt{2}$.

Answer:4 $\sqrt{2}$ (~5.66)

Q12. Section formula kis kaam?

- Step: Segment ko ratio me baant ne ke liye.

Answer:Point dividing in m1:m2

Q13. (-2,3),(4,3) midpoint?

- Step: $((-2+4)/2, (3+3)/2) = (1, 3)$.

Answer:(1, 3)

Q14. Q3 me signs?

- Step: Teesra quadrant.

Answer:(-, -)

Q15. (0,0),(0,7) doori?

- Step: $\sqrt{0 + 49} = 7$.

Answer:7

Q16. Midpoint(4,5), end(2,3) $\rightarrow$ doosra?

- Step: $(2+x)/2=4 \rightarrow x=6$; $(3+y)/2=5 \rightarrow y=7$.

Answer:(6, 7)

Q17. (3,0),(0,4) doori?

- Step: $\sqrt{9+16} = \sqrt{25}$.

Answer:5

Q18. A(2,3),B(8,3) ko 1:1?

- Step: 1:1 matlab midpoint = $((2+8)/2, 3)$.

Answer:(5, 3)

Q19. (7,-2) quadrant?

- Step: x+, y- $\rightarrow$ (+,-).

Answer:Q4

Q20. Collinear points?

- Step: Ek hi seedhi line par hone wale points.

Answer: Same line par points